PORTFOLIO

JUNHEE KIM, MACHINE LEARNING ENGINEER

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Building reliable ML systems for sparse, noisy, and shifting data

1. Statistical rigor and reliability

✓ **I model uncertainty and validate performance under noise, imbalance, and distribution shifts.**

2. Problem definition and technical judgment

✓ **I translate broad goals into measurable ML objectives, data requirements, and evaluation criteria.**

3. End-to-end implementation

✓ **I connect preprocessing, modeling, evaluation, and deployment into working systems.**

Selected Achievements

2026	Inha AI Challenge, Award of Excellence
2025	Inha AI Challenge, Grand Award
2025	LG Aimers 6, national finalist, 7th place
2025	Applied Sciences journal publication
2023	ITS Korea, Best Paper Award

Technical Skills

Python, R, SQL

PyTorch, TensorFlow, JAX

scikit-learn, Pandas, SciPy

Git, Docker, Airflow

HOW I OWN ML PROJECTS

1. Define the problem

I turn broad goals into explicit ML tasks and constraints.

- Define the target before choosing a model
- Separate in-scope decisions from assumptions
- Frame data scarcity and domain shift as testable problems

2. Design data and metrics

I let the data determine the modeling and evaluation plan.

- Inspect imbalance, noise, and distribution changes
- Select offline metrics that match the target
- Use cross-domain validation, LOOCV, calibration, and Brier score

3. Build and test

I compare technical choices through controlled experiments.

- Contrastive learning, Gaussian processes, and Bayesian optimization
- Benchmark against model and search baselines
- Use repeated experiments and ablation studies

4. Deliver the system

I implement the pipeline beyond a notebook.

- Sensor collection, preprocessing, training, and device deployment
- C++ export and BLE streaming on edge hardware
- Led schedules, roles, presentations, and final deliverables

01. Data-Scarce Anomaly Detection with Bayesian Optimization

Research paper: One-Class Classification Using Bayesian Optimization (under revision)

Data Scarcity | Bayesian Optimization | One-Class Learning | Model Selection

1. Objective

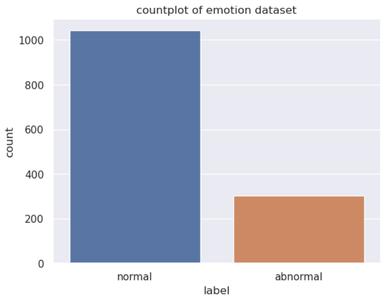
Detect abnormal patterns when labeled failures are scarce, while reducing the cost of tuning expensive one-class models.

2. Constraints

- Abnormal examples were too rare for standard binary classification.
- Deep-SVDD and OC-SVM made exhaustive hyperparameter search computationally expensive.

3. Technical approach

- Fit a Gaussian-process surrogate to validation performance.
- Use an upper confidence bound acquisition function to select the next configuration.
- Benchmark against grid search, random search, and Hyperband on synthetic data and a 24:1 imbalanced wildfire dataset.
- Compare both predictive performance and the number of model evaluations.

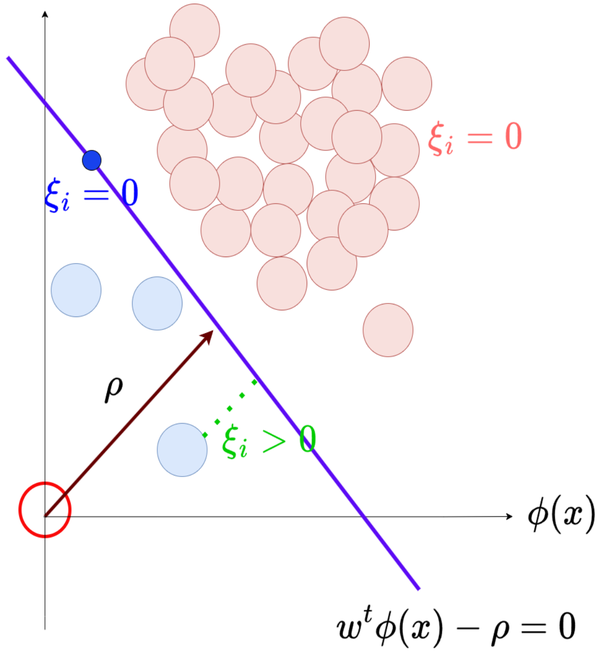


Imbalanced data

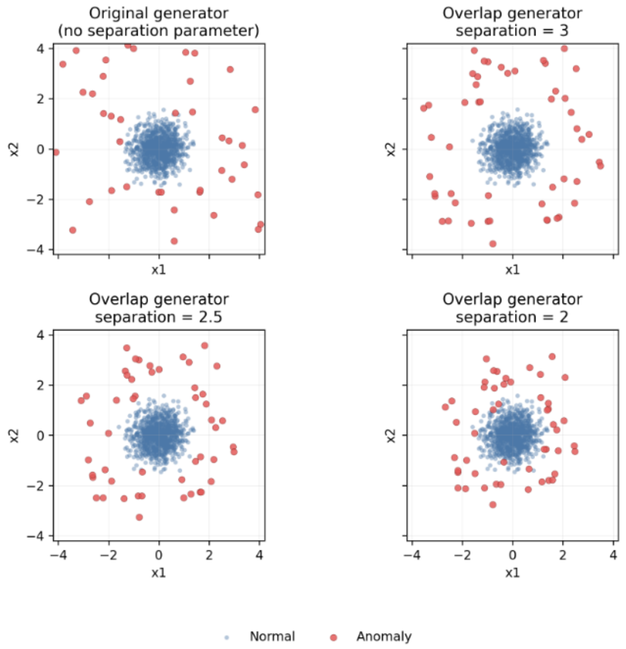
Metric	GC	RC	BO	HB	BOHB
Recall	0.4800	0.4350	0.4700	0.6200	0.6650
F-1	0.6142	0.5592	0.6569	0.4563	0.4155
AUC	0.6929	0.6407	0.7050	0.6407	0.6224

F1 and AUC comparison across search strategies

OC-SVM



Synthetic class-imbalance experiments



4. Relevance to Match Group AI

- Reached the best F1 and AUC range with fewer evaluations than exhaustive search strategies.
- The same framework supports cold-start, rare-event, and privacy-limited settings where labels are thin and each experiment is costly.

02. Contrastive Pretraining for Sparse Labels

Journal paper and Best Paper Award, ITS Korea

Contrastive Learning | Transfer Learning | Domain Shift | Time Series

1. Objective

Classify abnormal driving patterns from three-axis smartphone accelerometer data with only a small labeled set.

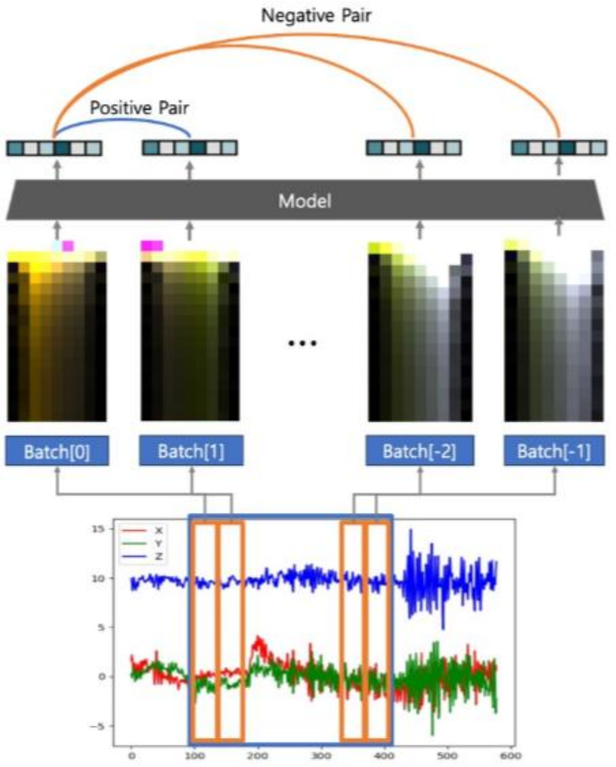
2. Constraints

- Manual labeling of driving behavior was expensive, leaving only a small supervised dataset.
- Vehicle-specific vibration patterns created a domain shift between training and deployment conditions.

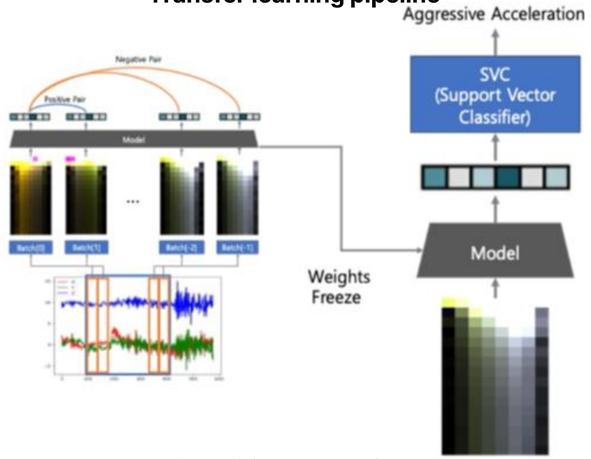
3. Technical approach

- Segment the sensor stream and apply STFT to preserve time-frequency structure.
- Construct positive and negative pairs from a time-consistency hypothesis.
- Pretrain a lightweight encoder with contrastive learning, freeze it as a feature extractor, and train an SVC with about 30 labels per class.
- Train and validate on different vehicles to test robustness to domain shift.

Time-consistency contrastive learning



Transfer learning pipeline



Classifier comparison

Model Name	Change Point Detection Result		Classification Result	
	Threshold	Precision	Precision	Recall
SqueezeNet	0.0	0	1.0	1.0
	0.4	0.75		
	0.6	0.69		
ShuffleNet	0.0	0	0.886	0.869
	0.4	0.73		
	0.6	0.61		
RegNet	0.0	1	0.956	0.951
	0.4	0.92		
	0.6	0.72		
MobileNet	0.0	0	0.039	0.197
	0.4	0.69		
	0.6	0.60		
EfficientNet	0.0	0	0.26	0.34
	0.4	0.67		
	0.6	0.76		
MnasNet	0.0	0	0.17	0.25
	0.4	0		
	0.6	0		

4. Relevance to Match Group AI

- Achieved precision 1.0 and recall 1.0 in the reported evaluation.
- Demonstrates a practical path from unlabeled behavioral logs to transferable representations when product labels are sparse or delayed.

03. Bayesian Deep Kernel Machines for Uncertainty-Aware Prediction

Published in Applied Sciences: Predicting Flatfish Growth in Aquaculture Using Bayesian Deep Kernel Machines

Nonlinear Prediction | Deep Kernel Machine | Gaussian Process | Uncertainty

1. Objective

Predict growth from multivariate environmental sensors and quantify predictive uncertainty for risk-sensitive decisions.

2. Constraints

- Environmental variables interacted through nonlinear and difficult-to-control effects.
- Standalone deep networks captured nonlinear patterns but did not expose how much confidence to place in each prediction.

3. Technical approach

- Combine a neural feature extractor with a Gaussian-process kernel in a Bayesian deep kernel machine.
- Evaluate the model with leave-one-out cross-validation on longitudinal sensor and growth data.
- Compare predictions with statistical and kernel baselines while retaining predictive intervals.

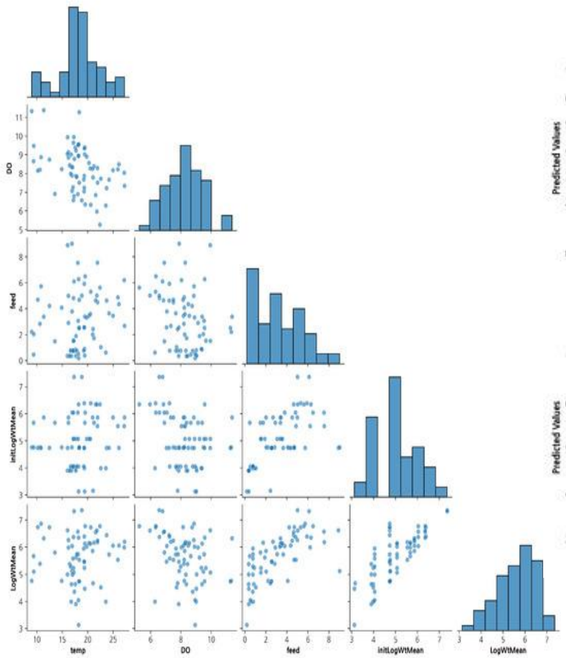
Model	MAE	MSE
KRR	1.1141	3.5665
BKMR	0.6977	0.9447
BDKMR (Equal)	0.2006	0.0721
BDKMR	**0.1895**	**0.0629**

Prediction error by model

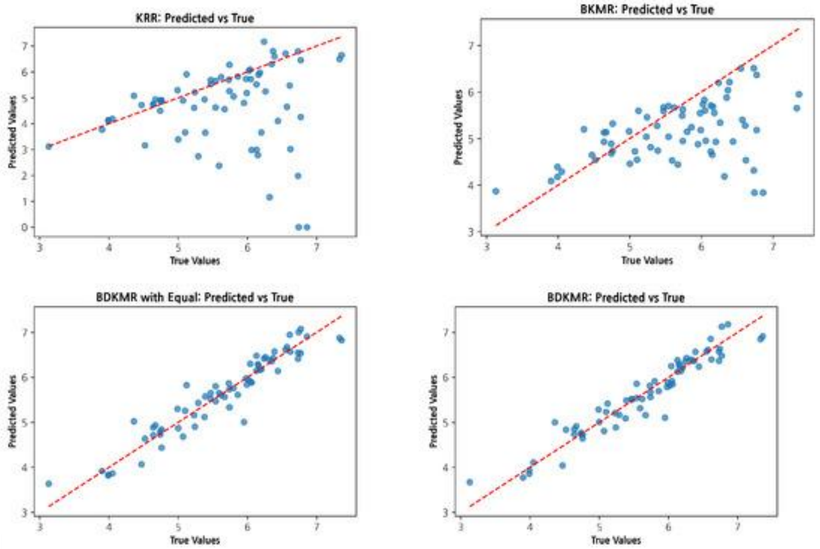
Model	Interpretability	Interaction Modeling	Uncertainty	Nonlinearity	Comp. Cost	Scalability
BKMR	Moderate	Strong	Yes	Strong	High	Moderate
BDKMR (ours)	Moderate	Strong	Yes	Very Strong	High	Moderate
Random Forest	Moderate	Limited	No	Moderate	Low	High
SVM (RBF kernel)	Low	Limited	No	Strong	Moderate	High

Comparison with nonlinear baselines

Feature interactions



Predicted and observed values across models



4. Relevance to Match Group AI

- Achieved MAE 0.1895 in the reported cross-validation and produced prediction intervals.
- Separating the point estimate from its uncertainty supports safer target setting, model evaluation, and downstream policy decisions.

04. Edge ML under a 256 KB Memory Limit

Edge Computing project, September to December 2023

Edge ML | Anomaly Detection | Arduino | End-to-End Pipeline | Deployment

1. Objective

Build a real-time system that collects vibration data, classifies equipment condition, and runs entirely on an edge device.

2. Constraints

- Cloud inference introduced latency and data-transfer concerns.
- Equipment-specific vibration patterns limited rule-based detection.
- The Arduino Nano 33 BLE provided only 256 KB of RAM, and no suitable sensor enclosure existed.

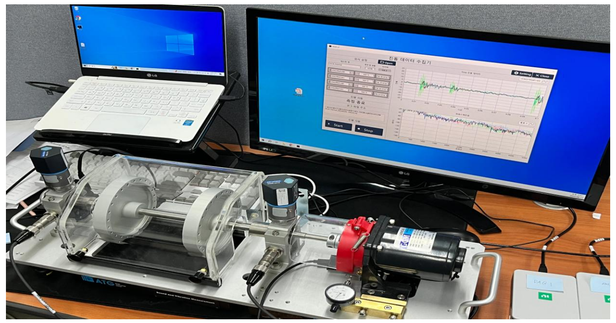
3. Technical approach

- Designed and 3D-printed a custom enclosure after learning the modeling workflow.
- Converted three-axis acceleration into frequency features with FFT and extracted RMS, skewness, kurtosis, and peak energy.
- Trained a lightweight 1D CNN, applied INT8 quantization, exported a C++ library, and deployed it through Arduino IDE.

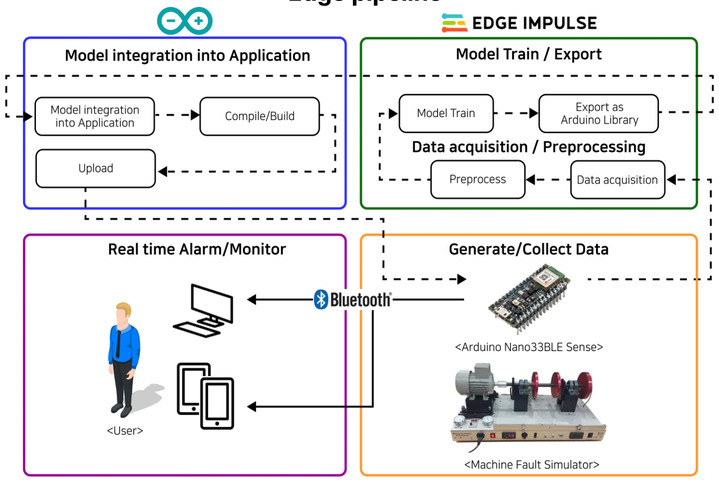
4. Deployment result

- Classified normal, imbalance, and misalignment states with 96.8% accuracy.
- Completed on-device inference without cloud communication.
- Validated the full path from sensor collection and preprocessing to model deployment and live output.

Data collection setup



Edge pipeline



5. ML systems value

- Shows ownership beyond model training, including data collection, hardware constraints, model compression, C++ integration, and runtime verification.

- The same discipline applies to production training and serving pipelines where latency, cost, and reliability shape model choices.

SELECTED RESEARCH AND ONGOING WORK

Generative models, uncertainty, and decision-focused machine learning

1. Domain-informed composite-kernel Gaussian process for molecular toxicity prediction
Designed a composite kernel using molecular similarity and domain features. The work focuses on predictive probability, uncertainty, and conformal validation and has been submitted to the Journal of Cheminformatics.

2. Language-guided image colorization, 2025 Inha AI Challenge Grand Award
Fine-tuned an LCAD diffusion model under limited compute and improved outputs through post-processing.

3. LG Aimers 6 clinical prediction, national finalist, 7th place
Built a CatBoost and LightGBM stacking ensemble and engineered clinical features to optimize a complex evaluation metric.

4. Bayesian Kernel Ridge on GAT Embeddings for Molecular Prediction
Combined graph-attention embeddings with Bayesian regression. Presented at BayesComp 2025 in Singapore.

5. Current research
Studying graph generative models for structural changes under radiation and Predictive Score Posterior methods for robust inference under model misspecification.